Flood Prediction and Alert System

Abstract

Floods are among the most destructive natural disasters, causing severe damage to life and property. This research presents a flood prediction and alert system that utilises data analysis and machine learning techniques to forecast flood occurrences and provide early warnings. The system aims to improve disaster preparedness and minimise risks through timely alerts.

Keywords:

Flood Prediction, Machine Learning, Wireless Sensor Network (WSN), Real-Time Monitoring, Disaster Alert System, Hydrological Modeling, Risk Assessment, HTML, CSS, Web Dashboard, Environmental Sensors

1 Introduction

Flood monitoring and alert system using machine learning is a project to collect flood data in all regions. The project will develop a system that uses a wireless sensor network which consists of sensors and transceivers to transmit data and predict floods. Floods are one of the most devastating natural disasters, causing significant loss of life and property worldwide. Rapid urbanisation, climate change, and unpredictable rainfall patterns have increased the frequency and severity of floods. In many regions, the lack of timely warning systems leads to severe damage and delayed evaluation. Traditional flood monitoring systems rely mainly on manual observation and limited sensor-based techniques, which may not provide accurate real-time predictions. Recent research has explored the use of Wireless Sensor Networks (WSN), Internet of Things (IoT), and machine learning models for real-time flood monitoring and early warning systems. This project proposes a Flood Prediction and Alert System that utilises historical rainfall and river water-level data to predict flood occurrence using machine learning algorithms. When the predicted flood probability exceeds a predefined threshold, the system automatically generates alert notifications to warn residents in advance. The proposed system aims to improve prediction accuracy and provide timely alerts to minimise disaster impact. According to the survey [1], the flood monitoring and alerting system makes a substantial difference in helping residents become aware of upcoming disasters. Another study by [2] highlights that testing and validation are essential to ensure system accuracy, reliability, and usability. This process encompasses various levels and types of testing, including unit testing, integration testing, system testing, and user acceptance testing (UAT).

2 Literature Review

Our flood prediction system consists of three main components working together:

- 1. Hydrological Model: Analyzes rainfall and river data to predict floods.
- 2. Flask Backend: Handles backend processing and calculations.
- 3. HTML/CSS/JavaScript Frontend: Provides the user interface where people can view predictions and receive alerts.

The system operates in a simple cycle. First, it collects weather and river data from monitoring stations. Then, the hydrological model and linear regression

process this data and calculate flood risk by evaluating or calculating the rainfall. The Flask backend manages this calculation and stores results in a database. Finally, the HTML/CSS/JavaScript frontend displays the predictions on an easy-to-use dashboard accessible through any web browser. In this project, we also included the capability to send alert messages and suggest safe routes via map directions, which is a key new feature.

3 Methodology

Methodology The methodology for developing the Flood Prediction and Alert System was designed to ensure accurate, real-time flood risk assessment and effective communication with end-users. The process encompasses data acquisition, backend processing, alert generation, and user interaction through a modern web interface. The steps are as follows:

1. Data Acquisition The system begins by collecting real-time and historical environmental data from a network of distributed monitoring stations. These stations measure parameters such as rainfall, river water levels, and temperature. Data is transmitted wirelessly to a centralised server using robust communication protocols to ensure timely and reliable delivery, even under adverse weather conditions. Rigorous data validation and preprocessing routines are implemented to address missing values, remove anomalies, and ensure the integrity and consistency of the dataset.

2. Backend Processing Upon receiving and cleaning the data, the backend—implemented using the Flask framework—serves as the system's processing hub. The backend is responsible for:

Storing raw and processed data in a structured database (PostgreSQL). Managing data flow between the acquisition layer, predictive models, and the user interface. Periodically invoking predictive algorithms (hydrological and machine learning models) to analyze the latest available data and compute flood risk levels for different locations. Logging all predictions and alert dispatches for future audit and evaluation.

3. Alert Generation and Safe Route Recommendation When the computed flood risk for a location exceeds predefined safety thresholds, the backend automatically initiates the alerting process. The alert subsystem:

Sends notifications via SMS and email to registered users in the affected regions. Updates alert status in the database for transparency and tracking. Generates and delivers safe route recommendations using integrated map services, helping users evacuate efficiently and avoid high-risk flood zones.

4. User Interface and Interaction The system's frontend is developed using HTML, CSS, and JavaScript, ensuring broad accessibility and responsiveness across devices. Through the web-based dashboard, users can:

View current flood risk levels, updated in real time. Explore interactive maps highlighting flood-prone areas and safe evacuation routes. Consult historical trends in rainfall and water levels. Subscribe to or manage their alert preferences for proactive notification. Access guidance and recommendations during emergency situations.

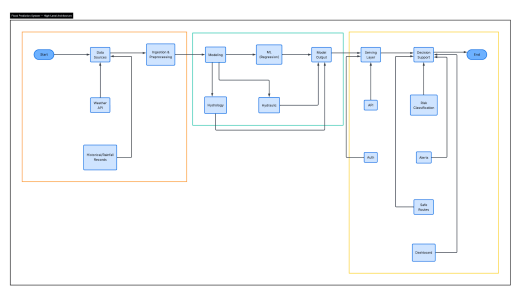

Figure 1: Overview

5. System Integration and Deployment All components—data acquisition, backend processing, and frontend—are integrated via RESTful APIs, enabling seamless communication and modular development. The system is containerized using Docker, allowing for straightforward deployment, scalability, and maintenance. Continuous monitoring and logging are employed to ensure high system uptime and quick recovery from errors.

6. Evaluation and Continuous Improvement Sys-

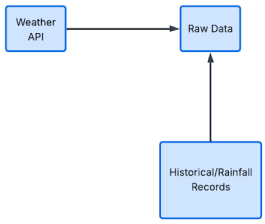

Figure 2: Data collection

tem performance is regularly evaluated using key metrics such as prediction accuracy, lead time, response time, and overall reliability. Feedback from users and stakeholders is incorporated to refine both the technical components and the user experience. The modular design facilitates ongoing updates, including the integration of new data sources, model enhancements, and additional alerting features

4 Data Collection

We gathered historical data from the local meteorological department and river monitoring stations. The dataset includes rainfall measurements, river water levels, and temperature readings from 2015 to 2024. This nine-year period covers several major flood events in our study area, providing sufficient data to train and test our model. The raw data had some gaps where sensors failed or readings were missed. We filled these gaps using interpolation methods based on nearby stations. Overall, about 3% of the data needed this treatment. We also removed obvious errors, such as negative rainfall values or physically impossible water levels from sensor malfunctions.

5 Hydrological modeling

Hydrological Modelling:

We chose the Rational Method for flood prediction because it works well for our urban catchment area and doesn't require complex calculations. This method is well-established in hydrology and has been used successfully in similar regions. The core equa-

Figure 3: Hydrology model

tion predicts peak discharge:

$Q = C \times I \times A$

Where:

- Q is the flood discharge in cubic meters per second,
- C is the runoff coefficient (how much rain becomes runoff),
- I is the intensity of the rainfall in millimetres per hour,
- A is the catchment area in square kilometres.

Runoff coefficients were determined on the basis of land use maps and soil type data. Urban areas with concrete and asphalt have higher coefficients (around 0.8) because water cannot soak into the ground. Green spaces and parks have lower values (around 0.3) because the soil absorbs more water. To account for the time it takes for water to flow through the catchment, we calculated the time of concentration using Kirpich's formula. This tells us how long after rainfall starts we should expect peak flow in the river. We calibrated the model using known flood events from 2015–2020 and tested it on events from 2021–2024 that the model had never seen before.

6 Backend Implementation with Flask

The Flask backend serves as the brain of our system. We chose Flask because it's lightweight, easy to deploy, and works well with Python's scientific libraries needed for hydrological calculations.

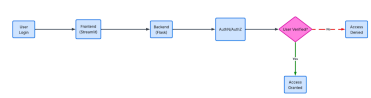

Figure 4: Backend Flow

Key backend functions include:

- The prediction module loads current weather data and feeds it into our hydrological model, determining the flood risk level—low, medium, high, or critical.

- The database module stores all predictions and historical data in a PostgreSQL database to track trends over time.

- API endpoints allow the frontend to access information. The main endpoint is /api/predict, which takes latitude and longitude as input and returns the flood prediction for that location. Another endpoint, /api/alerts, handles sending notifications when flood risk is too high.

- The alert system checks predictions every hour. If the risk exceeds the threshold, it automatically sends SMS messages to registered users in affected areas using the Twilio API. Email alerts are sent through SendGrid for users who prefer that option.

7 Frontend Development with HTML & CSS

To maximize accessibility and platform independence, the user interface is developed using HTML, CSS, and JavaScript. Users can access the dashboard from any device with a web browser. The main dashboard displays current flood risk for any location, using a color-coded indicator (green for low, yellow for moderate, orange for high, red for critical). Users can enter their location or select it on an interactive map, which updates in real time. Other features of the frontend include:

- An interactive flood risk map for regional or neighbourhood views.

- Alert subscription forms for SMS and email notifications.

- Historical charts showing trends in flood risk and rainfall.

- Safe route recommendations, guiding users along evacuation paths that avoid high-risk zones, displayed directly on the map.

The frontend communicates with the Flask backend via RESTful API endpoints, allowing seamless real-time updates. Responsive design ensures usability across desktops, tablets, and smartphones.

8 System Integration

Connecting all three components required careful planning. When a user requests a prediction via the frontend, it sends an HTTP request to Flask's API endpoint. Flask retrieves current weather data, runs it through the hydrological model, and sends the result back to the frontend. This process takes only a few seconds. The entire system is deployed using Docker containers—Flask runs in one, the database in another, and the frontend in a third. This setup makes it easy to scale and simplifies deployment. The database stores sensor data, prediction history, and user alert subscriptions.

Figure 5: System Integration Flowchart

9 Model Training and Validation

The dataset was split into training and testing portions. The training set included data from 2015 to 2020 for calibrating model parameters. The model was validated on flood events from 2021 to 2024, which it had not previously seen. Accuracy was measured by comparing predicted flood levels to observed outcomes. Results showed the model performs well for moderate floods, as the training data contained more examples of these events. Extreme events were predicted with less accuracy, a limitation noted for future improvement.

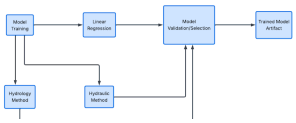

Figure 6: Model and Validation

10 Performance Metrics

The system was evaluated using several metrics:

- Prediction Accuracy: Percentage of forecasts correctly identifying flood occurrence.
- Lead Time: Hours of advance warning provided before flood conditions occur.
- Response Time: How quickly the system generates predictions after user requests.
- Nash-Sutcliffe Efficiency: Values above 0.7 indicate good hydrological model performance.
- Root Mean Square Error: Measures how far predictions deviate from reality.
- System Uptime: Ensures the warning system is available when needed.
- When it reaches the threshold value , the alert triggered.

11 Conclusion

This research presents a comprehensive approach to flood prediction and alerting by integrating environmental sensors, hydrological modeling, and web-based technologies. The implementation of a wireless sensor network ensures continuous, real-time data collection, while the use of machine learning enhances the system's prediction accuracy for various flood scenarios. By adopting an HTML- and CSS-based frontend, the system remains user-friendly and accessible to a broad audience. Automated alert mechanisms further contribute to community preparedness by providing timely warnings and safe evacuation guidance. Overall, the proposed system demonstrates significant potential to reduce the impact of floods through early detection and effective communication. Continued refinement, including the incorporation of more diverse data sources and advanced predictive models, can further improve system performance and reliability in the future.

12 Advantages

The proposed flood prediction and alert system offers several notable advantages:

Real-Time Monitoring: Continuous data collection and analysis allow for immediate detection of flood risks as soon as conditions change. Automated Alerts: The system provides timely SMS and email notifications, ensuring that residents and authorities receive warnings without delay. User Accessibility: With a web-based HTML/CSS/JavaScript frontend, users can access flood information and safe route guidance from any device with an internet connection. Data-Driven Decision Making: Integration of machine learning and hydrological models enhances prediction accuracy, supporting more informed emergency response. Scalability and Flexibility: Modular architecture and containerized deployment (using Docker) make the system easy to scale and adapt to different regions or expand with new features. Historical Analysis: The system maintains a comprehensive record of past events, supporting trend analysis and ongoing model improvement.

13 Disadvantages

Disadvantages While the system brings significant benefits, some limitations remain:

Sensor Reliability: The accuracy of predictions depends on the continuous operation and maintenance of environmental sensors, which may be affected by harsh weather or technical failures. Data Quality: Errors or gaps in sensor data can reduce model effectiveness, despite preprocessing and interpolation efforts. Limited Prediction for Extreme Events: The machine learning model may struggle with rare or unprecedented flood scenarios due to insufficient historical examples. Dependence on Communication In-

frastructure: Real-time alerts and data transfer require stable internet and mobile networks, which may be disrupted during severe weather conditions. User Adoption: Ensuring widespread use of the system and understanding among all community members can be challenging, especially in areas with limited digital literacy.

14 Future enhancement

Future Enhancements To further improve the effectiveness and resilience of the flood prediction and alert system, the following enhancements are proposed:

Integration of Satellite and Drone Data: Incorporating remote sensing data can expand monitoring coverage and provide additional context for flood prediction, especially in remote or inaccessible areas. Advanced Machine Learning Models: Adopting deep learning techniques, such as recurrent neural networks (RNNs) or long short-term memory (LSTM) networks, could improve the system's ability to recognize complex temporal patterns and enhance forecasting accuracy. Multilingual and Voice-Based Alerts: Expanding alert delivery to include multiple languages and voice calls can make the system more inclusive, particularly for users who may not read text messages or emails. Mobile Application Development: Creating a dedicated mobile app could offer push notifications, offline access to safe routes, and personalized risk dashboards. Community Feedback Mechanism: Allowing users to report local flood conditions or false alarms can help validate and refine the system, making it more adaptive to real-world scenarios. Integration with Emergency Services: Direct collaboration with local authorities and emergency response teams can streamline evacuation coordination and resource deployment during flood events.

15 Citation

Bibtex:

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[1] M. J. Subashini, R. Sudarmani, S. Gobika and R. Varshini, "Development of Smart Flood Monitoring and Early Warning System using Weather Forecasting Data and Wireless Sensor Networks-A Review," 2021 Third International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV), Tirunelveli, India, 2021, pp. 132-135, doi: 10.1109/ICICV50876.2021.9388418.

[2] NINYESIGA, M. A. (2025). A FLOOD EARLY WARNING SYSTEM (Doctoral dissertation, School of Computing and Engineering, Uganda Technology and Management University).